

An Elementary Proof that
 $48X^4 + 12X^2 + 1 = Y^2$ has only the Solutions $(X, Y) = (0, \pm 1)$

Theorem 1. *The only integer solutions of $48X^4 + 12X^2 + 1 = Y^2$ are $(X, Y) = (0, 1)$ and $(0, -1)$.*

Strategy. We reduce the quartic to the Pell equation $U^2 - 3V^2 = 1$, then run a two-stage infinite descent on the index of the Pell sequence, terminating in the auxiliary quartic $a^4 - 3b^4 = 1$. That equation is then ruled out by a classical 1914 theorem of H. C. Pocklington (References, p. 4), which finishes the proof.

Proof. By symmetry we may take $Y \geq 0$, and we argue by contradiction: suppose $X \neq 0$.

Step 1: Reduction to a Pell equation.

Multiplying the defining equation by 4 and comparing with $3(8X^2 + 1)^2 = 192X^4 + 48X^2 + 3$ gives

$$4Y^2 - 3(8X^2 + 1)^2 = (192X^4 + 48X^2 + 4) - (192X^4 + 48X^2 + 3) = 1. \quad (1)$$

So, setting

$$U := 2Y, \quad V := 8X^2 + 1,$$

the pair (U, V) satisfies the Pell equation

$$U^2 - 3V^2 = 1. \quad (2)$$

Step 2: The Pell sequence.

Every solution of (2) in non-negative integers is (U_n, V_n) , $n \geq 0$, where

$$U_n + V_n\sqrt{3} = (2 + \sqrt{3})^n, \quad U_0 = 1, \quad V_0 = 0, \quad U_{n+1} = 2U_n + 3V_n, \quad V_{n+1} = U_n + 2V_n.$$

Both (U_n) and (V_n) satisfy the second-order recurrence $a_{n+2} = 4a_{n+1} - a_n$, coming from the minimal polynomial $t^2 - 4t + 1$ of $2 + \sqrt{3}$. Since $U_n^2 - 3V_n^2 = 1$ for every n , any common divisor of U_n, V_n divides 1:

$$\gcd(U_n, V_n) = 1 \quad \text{for all } n \geq 0. \quad (3)$$

Step 3: Which index can occur.

We need $V = 8X^2 + 1 \equiv 1 \pmod{8}$. Reducing $V_{n+2} = 4V_{n+1} - V_n$ modulo 8 and starting from $(V_0, V_1) \equiv (0, 1) \pmod{8}$ gives the sequence

$$V_n \pmod{8} = 0, 1, 4, 7, 0, 1, 4, 7, \dots$$

which is exactly periodic with period 4 (the pair $(V_4, V_5) \equiv (0, 1) \equiv (V_0, V_1) \pmod{8}$, so the recurrence repeats). Hence $V_n \equiv 1 \pmod{8}$ forces

$$n \equiv 1 \pmod{4}, \quad \text{say } n = 2k + 1 \text{ with } k \text{ even.}$$

Step 4: An identity for V_{2k+1} .

Squaring $(U_k + V_k\sqrt{3})^2 = (U_k^2 + 3V_k^2) + 2U_kV_k\sqrt{3}$ and multiplying by $(2 + \sqrt{3})$ gives $U_{2k+1} + V_{2k+1}\sqrt{3}$, whose $\sqrt{3}$ -coefficient is

$$V_{2k+1} = U_k^2 + 4U_kV_k + 3V_k^2. \quad (4)$$

Using $U_k^2 = 1 + 3V_k^2$ (the Pell relation at index k) to eliminate U_k^2 ,

$$V_{2k+1} = 1 + 4U_kV_k + 6V_k^2 = 1 + 2V_k(2U_k + 3V_k). \quad (5)$$

Combining (5) with $8X^2 = V_{2k+1} - 1$ from Step 3,

$$4X^2 = V_k(2U_k + 3V_k). \quad (6)$$

Step 5: Parity, and an exact common factor of 2.

From the period-4 pattern of Step 3, $V_k \bmod 8 \in \{0, 4\}$ for even k , so

$$4 \mid V_k. \quad (7)$$

Since $U_k^2 = 1 + 3V_k^2$ and V_k is even, U_k^2 is odd, so

$$U_k \text{ is odd.} \quad (8)$$

Consequently $2U_k + 3V_k$ is even (odd $\cdot 2$ plus even $\cdot 3$), and by (7) so is V_k ; thus $2 \mid \gcd(V_k, 2U_k + 3V_k)$. Conversely, any common divisor d of V_k and $2U_k + 3V_k$ divides $(2U_k + 3V_k) - 3V_k = 2U_k$, so $d \mid \gcd(2U_k, V_k)$; as $\gcd(U_k, V_k) = 1$ by (3), this forces $d \mid 2$. Hence

$$\gcd(V_k, 2U_k + 3V_k) = 2 \quad \text{exactly.} \quad (9)$$

Step 6: Extracting squares.

Divide (6) by the common factor 2 found in (9):

$$X^2 = \frac{V_k}{2} \cdot \left(U_k + \frac{3V_k}{2} \right),$$

a product of two *coprime* non-negative integers (their gcd was exactly removed). Coprime factors of a perfect square are themselves perfect squares, so there exist non-negative integers r, s with

$$V_k = 2r^2, \quad U_k + \frac{3}{2}V_k = s^2. \quad (10)$$

Step 7: The second descent, via $k = 2m$.

Since k is even, write $k = 2m$. Squaring $(U_m + V_m\sqrt{3})^2 = (U_m^2 + 3V_m^2) + 2U_mV_m\sqrt{3}$ gives the doubling identity

$$V_k = V_{2m} = 2U_mV_m. \quad (11)$$

Comparing (11) with $V_k = 2r^2$ from (10) gives $U_mV_m = r^2$; since $\gcd(U_m, V_m) = 1$ by (3), each factor is again a perfect square:

$$U_m = a^2, \quad V_m = b^2 \quad \text{for some non-negative integers } a, b. \quad (12)$$

Step 8: The auxiliary quartic.

Substituting (12) into the Pell relation $U_m^2 - 3V_m^2 = 1$ at index m ,

$$a^4 - 3b^4 = 1. \quad (13)$$

Because we assumed $X \neq 0$, we have $n \geq 5$ (the least admissible index beyond $n = 1$), so $k = (n - 1)/2 \geq 2$ and $m = k/2 \geq 1$. Then $V_m \geq V_1 = 1 > 0$ and $U_m \geq U_1 = 2 > 0$, so (12) forces $a, b \geq 1$: equation (13) has a solution in *positive* integers.

Step 9: Closing the descent.

Lemma 1 (Pocklington, 1914 — References, p. 4). *The equation $x^4 - 3y^4 = z^2$ has no solution in positive integers x, y, z .*

Taking $z = 1$ shows $a^4 - 3b^4 = 1$ has no solution with $a, b > 0$, contradicting the conclusion of Step 8. Hence the assumption $X \neq 0$ is false.

Step 10: The trivial case.

If $X = 0$ then $Y^2 = 1$, so $Y = \pm 1$; direct substitution confirms $48 \cdot 0^4 + 12 \cdot 0^2 + 1 = 1 = (\pm 1)^2$.

Combining Steps 1–10, the only integer solutions are $(X, Y) = (0, \pm 1)$. $\square$

Remark (shape of the descent). The proof compresses to the chain

$$48X^4 + 12X^2 + 1 = Y^2 \implies U^2 - 3V^2 = 1 \implies U_m = a^2, V_m = b^2 \implies a^4 - 3b^4 = 1,$$

after which Pocklington’s classical result finishes it. The genuinely load-bearing algebraic fact, isolated in Steps 4–7, is that the index of a Pell solution of the required residue class ($n \equiv 1 \pmod{4}$) can itself be halved twice, and each halving forces a further pair of coprime squares.

Notes on this write-up. This version fixes three issues present in an earlier draft of the same argument:

1. Step 5–6 make the exact common factor explicit ($\gcd(V_k, 2U_k + 3V_k) = 2$, proved from $\gcd(U_k, V_k) = 1$ and parity) rather than asserting the square-extraction outright; this also removes a factor-of-2 inconsistency in how r was defined in the earlier draft ($V_k = 2r^2$ here, throughout).
2. The mod-8 periodicity claimed in Step 3 (and the resulting $4 \mid V_k, U_k$ odd used in Step 5) is derived from the linear recurrence rather than only stated.
3. The citation in Step 9 has been checked against the primary source (References, p. 4): volume, year, and page range are confirmed to match exactly. What was *not* independently re-derived is the verbatim statement of Pocklington’s theorem from the original 14-page paper itself, which was accessible only via its table of contents and secondary citations in this session – so Step 9 is a well-attested classical citation, not a re-proof from the primary text.

References

1. H. C. Pocklington, *Some Diophantine Impossibilities*, Proceedings of the Cambridge Philosophical Society, Vol. 17 (1914), pp. 108–121. Primary source (full scanned volume, 1913–1916):
<https://archive.org/details/proceedingsofcam1718191316camb>
(The table of contents of this volume lists the paper as beginning on p. 108, with the next article starting p. 122, confirming the pp. 108–121 range. The paper is cited as the standard source for infinite-descent proofs that various quartic forms are never perfect squares.)
2. Alan Baker, *A Comprehensive Course in Number Theory*, Cambridge University Press, 2012, ISBN 978-1-107-60379-0. Book page:
<https://www.cambridge.org/core/books/comprehensive-course-in-number-theory/6E56BE58DA51F3223AB947279F250D61>
Chapter 8, “Diophantine equations”:
<https://www.cambridge.org/core/books/abs/comprehensive-course-in-number-theory/diophantine-equations/D8801805AEEF572F4E155D2A1340756C>